

JOINT TEMPORAL PATTERNS OF EATING AND 24-HOUR MOVEMENT BEHAVIOURS: ASSOCIATIONS WITH ADIPOSITY AND BLOOD PRESSURE IN AUSTRALIAN ADULTS

B. Jima¹, S. McNaughton^{1,2}, D. Dunstan^{1,3}, A. Kouzani⁴, R. Leech¹

¹ Institute for Physical Activity and Nutrition, Deakin University

² Health and Well-Being Centre for Research Innovation, University of Queensland

³ Baker Heart & Diabetes Institute, Melbourne

⁴ School of Engineering, Deakin University

Email: s223923898@deakin.edu.au

INTRODUCTION

The timing and amount of energy intake (EI), alongside the timing and duration of 24-hour movement behaviours (physical activity [PA], sedentary time [ST] and sleep duration [SD]), vary across the day and may interact to shape cardiometabolic health. However, most research has focussed on the temporal patterning of individual behaviours [1-4]; the joint temporal patterning of eating and movement behaviours has rarely been examined [5]. This study aimed to identify joint temporal patterns of eating and 24-hour movement behaviours and examine their associations with adiposity and blood pressure in Australian adults.

METHODS

This cross-sectional study analysed data from 142 adults aged 18-65 years in Victoria, Australia, who participated in the “EveryDayLife” survey (2017–2020). Participants completed a smartphone-based food diary and wore a SenseWear activity monitor armband for up to seven days. Diet quality was assessed using dietary guideline index. Multivariate dynamic time warping-based clustering was applied to hourly time-series data comprising EI and the duration of PA, ST and SD in each hour of the day, averaged across all valid record days. Multivariable linear and logistic regression examined associations with adiposity (body mass index [BMI], waist circumference [WC], body fat percentage [BF%], fat and fat-free mass indices, visceral adipose tissue [VAT]), blood pressure, and the prevalence of obesity, central obesity and hypertension, adjusting for relevant covariates.

RESULTS

Three clusters were identified. Clusters 1 (n = 26) and 2 (n = 105) showed comparable behavioural timing, with EI peaks at midday and early evening and sleep between 22:00 and 06:00 h but contrasting daily PA volumes (total PA 8.2 versus 5.1 h/day; ST 7.8 versus 10.7 h/day, respectively). Cluster 3 (n = 11) was

characterised by a shift of EI, PA, ST, and SD into the evening and overnight period, with a third of daily EI consumed between 20:00 and 00:00 h and SD concentrated between 03:00 and 11:00 h, alongside the lowest daily PA (3.8 h/day) and highest ST (12.3 h/day). Total EI and overall diet quality were comparable across clusters. Relative to Cluster 1, participants in Cluster 3 had higher WC ($\beta = 16.3\%$; 95% CI: 3.2, 31.1; $p = 0.013$), BF% ($\beta = 7.9\%$; 95% CI: 1.5, 14.2; $p = 0.015$), and fat mass index ($\beta = 3.2$ unit; 95% CI: 0.3, 6.0; $p = 0.031$), and greater odds of central obesity (OR = 10.2; 95% CI: 1.9, 54.0; $p = 0.006$) and excess BF% (OR = 7.9; 95% CI: 1.5, 42.7; $p = 0.017$). BMI, VAT, and blood pressure were not significantly different across clusters ($p > 0.05$).

CONCLUSIONS

In conclusion, a pattern combining later timing of EI and movement behaviours with low daily PA and high ST was associated with greater adiposity. Examining these behaviours jointly may capture patterns not identifiable by individual behaviours alone.

ACKNOWLEDGMENTS

“EveryDayLife” survey was supported by an Australian Research Council Discovery Grant [DP170100544]. BRJ is supported by the Deakin University Postgraduate Research Scholarship (DUPR).

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